



**Shore**

**Year 11  
Mathematics  
Yearly Examination  
September 2013**

**General Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Answer Questions 1–5 on the Multiple Choice Answer Sheet provided
- Start each of Questions 6–10 in a new writing booklet
- In Questions 6–10, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Examination Number:

Set:

**Total marks – 80**

**Section I** Pages 2–3

**5 marks**

- Attempt Questions 1–5
- Each question is worth 1 mark
- Allow about 10 minutes for this section

**Section II** Pages 4–8

**75 marks**

- Attempt Questions 6–10
- Each question is worth 15 marks
- Allow about 1 hour and 50 minutes for this section

**Note: Any time you have remaining should be spent revising your answers.**

**Section I**

**5 marks**

**Attempt Questions 1–5**

**Allow about 10 minutes for this section**

Use the multiple choice answer sheet for Questions 1–5.

1 Which one of the following is equivalent to  $\frac{x}{8} - \frac{x+1}{4}$  ?

- (A)  $\frac{1-x}{8}$
- (B)  $\frac{-1-x}{8}$
- (C)  $\frac{2-x}{8}$
- (D)  $\frac{-2-x}{8}$

2 Which one of the following is equal to  $\frac{1+\sqrt{5}}{7-2\sqrt{5}}$  ?

- (A)  $\frac{5\sqrt{5}-3}{29}$
- (B)  $\frac{5\sqrt{5}-3}{39}$
- (C)  $\frac{9\sqrt{5}+17}{29}$
- (D)  $\frac{9\sqrt{5}+17}{39}$

**DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM**

- 3 The roots of a quadratic equation are  $-3$  and  $4$ .

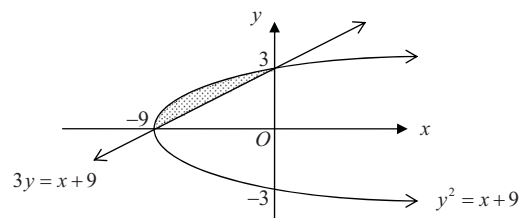
Which one of the following could be the quadratic equation?

- (A)  $x^2 - x - 12 = 0$   
 (B)  $x^2 + x - 12 = 0$   
 (C)  $x^2 - x + 12 = 0$   
 (D)  $x^2 + x + 12 = 0$

- 4 What are the solutions to  $\sqrt{3} \tan x = -1$  for  $0^\circ \leq x \leq 360^\circ$ ?

- (A)  $120^\circ$  and  $240^\circ$   
 (B)  $120^\circ$  and  $300^\circ$   
 (C)  $150^\circ$  and  $210^\circ$   
 (D)  $150^\circ$  and  $330^\circ$

- 5 The diagram shows the region enclosed by  $3y = x + 9$  and  $y^2 = x + 9$ .



Which one of the following pairs of inequalities describes the shaded region in the diagram?

- (A)  $3y \geq x + 9$  and  $y^2 \geq x + 9$   
 (B)  $3y \geq x + 9$  and  $y^2 \leq x + 9$   
 (C)  $3y \leq x + 9$  and  $y^2 \geq x + 9$   
 (D)  $3y \leq x + 9$  and  $y^2 \leq x + 9$

## Section II

**75 marks**

**Attempt Questions 6–10**

**Allow about 1 hour and 50 minutes for this section**

Start each of Questions 6–10 in a new writing booklet.

In Questions 6–10, your responses should include relevant mathematical reasoning and/or calculations.

**Question 6 (15 marks)** Use a SEPARATE writing booklet.

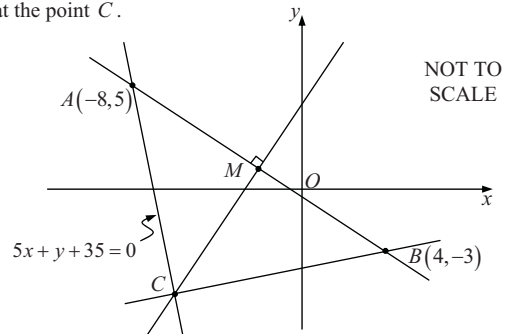
- (a) The selling price of a television is \$1056. This includes a 10% tax on the original price. Calculate the original price of the television. **2**
- (b) Find the perpendicular distance from the point  $(-2, 5)$  to the line  $4x + 3y = 12$ . **2**
- (c) Factorise fully  $x^2 - y^2 + 2x + 2y$ . **2**
- (d) Solve  $|3x - 1| < 7$ . **2**
- (e) Find the value of  $m$  if the line  $6x + my + 7 = 0$  is parallel to the line  $3x + 2y - 4 = 0$ . **2**
- (f) A regular polygon has an interior angle sum of  $3960^\circ$ . **2**  
 How many sides does the polygon have?
- (g) A function is given by  $f(x) = x^3 - 2x^2 + x$ . **3**  
 Find the values of  $x$  for which  $f'(x) = 0$ .

**Question 7 (15 marks)** Use a SEPARATE writing booklet.

(a) Differentiate  $(1-4x^2)^5$  with respect to  $x$ .

2

(b) The diagram shows the points  $A(-8,5)$  and  $B(4,-3)$ . The point  $M$  is the midpoint of the interval  $AB$ . The line  $5x + y + 35 = 0$  meets the perpendicular bisector of  $AB$  at the point  $C$ .



(i) Find the coordinates of  $M$ .

1

(ii) Find the gradient of  $AB$ .

1

(iii) Show that the equation of the perpendicular bisector of  $AB$  is  $3x - 2y + 8 = 0$ .

2

(iv) Show that the coordinates of the point  $C$  are  $(-6, -5)$ .

2

(v) Prove that  $\triangle ABC$  is right-angled at  $C$ .

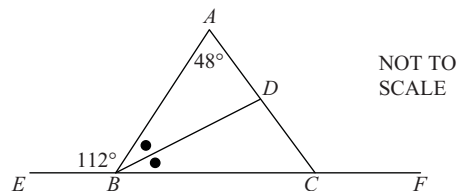
2

(vi) Find the area of  $\triangle ABC$ .

2

(c) In the diagram,  $\angle ABE = 112^\circ$ ,  $\angle BAC = 48^\circ$  and  $BD$  bisects  $\angle ABC$ .

3



Copy or trace the diagram into your writing booklet.

Find the size of  $\angle BDC$ , giving reasons for your answer.

- 5 -

**Question 8 (15 marks)** Use a SEPARATE writing booklet

(a) Simplify  $\tan \theta \sqrt{1 - \sin^2 \theta}$ .

2

(b) Find  $f'(x)$  if  $f(x) = \frac{1}{2x^4}$ .

2

Write your answer in simplest form with positive integer indices.

(c) Find  $\lim_{x \rightarrow 4} \left( \frac{x^2 + 3x - 4}{2x + 8} \right)$ .

2

(d) Consider the curve  $y = x^3 - 3x^2$ .

3

Find the equation of the tangent at the point on the curve where  $x = -1$ .

(e) For what values of  $k$  does  $3x^2 - 2kx + k = 0$  have real roots?

3

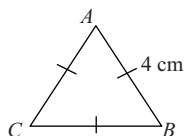
(f) Sketch the curve  $y = \frac{1}{x-3} + 1$  clearly labelling any intercepts with the coordinate axes and other important features.

3

- 6 -

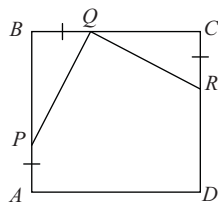
**Question 9 (15 marks)** Use a SEPARATE writing booklet.

- (a) Find the exact area of an equilateral triangle with side length 4 centimetres. 2



NOT TO  
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- (b) In the diagram,  $ABCD$  is a square. The points  $P$ ,  $Q$ , and  $R$  lie on  $AB$ ,  $BC$ , and  $CD$  respectively such that  $AP = BQ = CR$ .



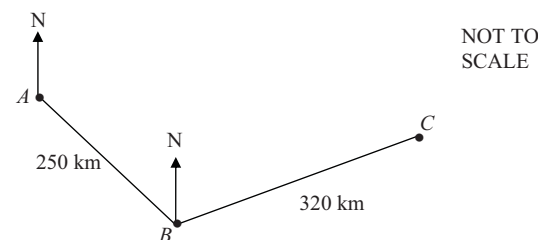
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Copy or trace the diagram into your writing booklet.

- (i) Prove that  $\triangle PBQ \cong \triangle QCR$ . 3
- (ii) Prove that  $\angle PQR$  is a right angle. 2
- (c) Solve  $2\sin^2 x = \sin x$  for  $0^\circ \leq x \leq 360^\circ$ . 2
- (d) Let  $\alpha$  and  $\beta$  be the roots of the quadratic equation  $4 - 8x - x^2 = 0$ .
- (i) Find  $\alpha + \beta$ . 1
- (ii) Find  $\alpha\beta$ . 1
- (iii) Find  $\alpha^2 + \beta^2$ . 2
- (iv) Find  $(\alpha - \beta)^2$ . 2

**Question 10 (15 marks)** Use a SEPARATE writing booklet.

- (a) Solve  $4 + 23x^2 - 6x^4 = 0$ . 2
- (b) Determine whether the function  $f(x) = 2x - x^3$  is even, odd or neither. 2
- (c) Write  $x^2$  in the form  $A(2-x)^2 + B(2-x) + C$ . 3
- (d) A ship sailed 250 kilometres from Port  $A$  on a bearing of  $153^\circ$  and arrived at Port  $B$  to pick up some passengers. It then progressed 320 kilometres to Port  $C$  on a bearing of  $071^\circ$ .



Copy or trace this diagram into your writing booklet.

- (i) Find the distance  $AC$  correct to the nearest kilometre. 3
- (ii) Hence, find the bearing of Port  $A$  from Port  $C$  correct to the nearest degree. 2
- (e) Find  $\frac{d}{dx} \left( \frac{4x^2}{\sqrt{3x-2}} \right)$ . 3

Write your answer in simplest form with positive integer indices.

**End of Paper**

Examination Number:

Set:

## Mathematics 2 Unit

### Section I – Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample:**  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9  
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐  
*correct* ↖

---

1 A ☐ B ☐ C ☐ D ☐

2 A ☐ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☐

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1. D                      4. D  
2. C                      5. B  
3. A

Question 6

(a) Original Price =  $\frac{\$1056}{1.1}$   
=  $\underline{\underline{\$960}}$

(b)  $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$   
=  $\frac{|4(-2) + 3(5) - 12|}{\sqrt{4^2 + 3^2}}$   
=  $\frac{|-5|}{5}$   
= 1 unit

(c)  $x^2 - y^2 + 2x + 2y$   
=  $(x-y)(x+y) + 2(x+y)$   
=  $(x+y)(x-y+2)$

(d)  $|3x-1| < 7$   
 $-7 < 3x-1 < 7$   
 $-6 < 3x < 8$   
 $\underline{\underline{-2 < x < 2\frac{2}{3}}}$

(e)  $-\frac{6}{m} = \frac{-3}{2}$   
 $-12 = -3m$   
 $\underline{\underline{m = 4}}$

(f)  $(n-2) \times 180^\circ = 3960^\circ$   
 $n-2 = 22$   
 $\underline{\underline{n = 24}}$

(g)  $3x^2 - 4x + 1 = 0$   
 $(3x-1)(x-1) = 0$   
 $x = \frac{1}{3} \text{ or } x = 1$

Question 7

(a)  $\frac{d}{dx} (1-4x^2)^5$   
=  $5(1-4x^2)^4 \times -8x$   
=  $\underline{\underline{-40x(1-4x^2)^4}}$

(b) (i)  $M = \left(\frac{-8+4}{2}, \frac{5+3}{2}\right)$   
=  $\underline{\underline{(-2, 1)}}$

(ii)  $m_{AB} = \frac{-3-5}{4+8}$   
=  $\frac{-8}{12}$   
=  $\underline{\underline{-\frac{2}{3}}}$

(iii)  $m_{AC} = \frac{3}{2}$      $M(-2, 1)$   
 $y-1 = \frac{3}{2}(x+2)$   
 $2y-2 = 3x+6$   
 $\underline{\underline{3x-2y+8=0}}$

(iv)  $5x + y + 35 = 0$  (1)  
 $3x - 2y + 8 = 0$  (2) } Solve:  
 $10x + 2y + 70 = 0$  (1e)  
 $3x - 2y + 8 = 0$  (2)  
(1e) + (2)     $13x + 78 = 0$

$x = -6$   
Sub  $x = -6$  into (1)  
 $5(-6) + y + 35 = 0$   
 $y + 5 = 0$

$y = -5$   
 $\therefore C$  is  $\underline{\underline{(-6, -5)}}$

(v)  $m_{AC} = \frac{-5-5}{-6+8} = \frac{-10}{2} = -5$   
 $m_{BC} = \frac{-5+3}{-6-4} = \frac{-2}{-10} = \frac{1}{5}$

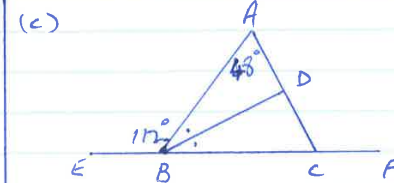
$m_{AC} \times m_{BC} = -5 \times \frac{1}{5} = -1$

$\therefore \underline{\underline{AC \perp BC}}$

(vi)  $AC = \sqrt{(-8+6)^2 + (5+5)^2}$   
=  $\sqrt{4+100}$   
=  $\sqrt{104}$

$BC = \sqrt{(4+6)^2 + (-5+3)^2}$   
=  $\sqrt{100+4}$   
=  $\sqrt{104}$

$\therefore \text{Area} = \frac{1}{2} \times \sqrt{104} \times \sqrt{104}$   
=  $\underline{\underline{52 \text{ units}^2}}$



$\angle ABC = 68^\circ$  straight  $\angle EBC$   
 $\angle ABD = 34^\circ$   $BD$  bisects  $\angle ABC$   
 $\angle BDC = 82^\circ$  Sum of 2 interior  $\angle$ s equals opp. ext.  $\angle$

Question 8

(a)  $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\sin \theta}{\sqrt{1-\sin^2 \theta}}$   
=  $\frac{\sin \theta}{\cos \theta} \times \sqrt{\cos^2 \theta}$   
=  $\frac{\sin \theta}{\cos \theta} \times \cos \theta$   
=  $\underline{\underline{\sin \theta}}$

(b)  $f(x) = \frac{1}{2} x^{-4}$   
 $f'(x) = -2x^{-5}$   
=  $\underline{\underline{-\frac{2}{x^5}}}$

(c)  $\lim_{x \rightarrow -4} \frac{x^2 + 3x - 4}{2x + 8}$   
=  $\lim_{x \rightarrow -4} \frac{(x+4)(x-1)}{2(x+4)}$   
=  $\underline{\underline{-\frac{5}{2}}}$

$$(d) \quad y = x^3 - 3x^2$$

$$y' = 3x^2 - 6x$$

$$\text{at } x = -1 \quad y = (-1)^3 - 3(-1)^2$$

$$= -1 - 3$$

$$= -4$$

$$m_T = 3(-1)^2 - 6(-1)$$

$$= 3 + 6$$

$$= 9$$

$$\therefore \text{equation is: } y + 4 = 9(x + 1)$$

$$y + 4 = 9x + 9$$

$$y = 9x + 5$$

$$(e) \quad \text{For real roots } \Delta \geq 0$$

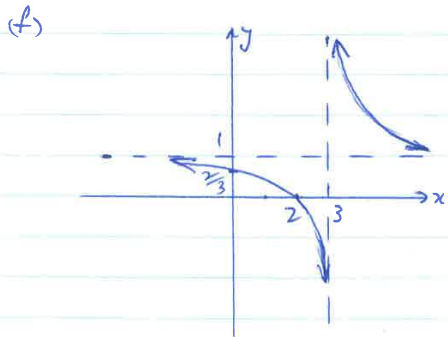
$$(-2k)^2 - 4 \times 3 \times k \geq 0$$

$$4k^2 - 12k \geq 0$$

$$4k(k - 3) \geq 0$$

$$\begin{array}{c} \vee \quad \times \quad \vee \\ 0 \quad \quad 3 \end{array}$$

$$\therefore \underline{k \leq 0 \text{ or } k \geq 3}$$

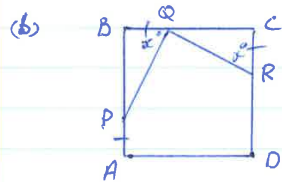


### Question 9

$$(a) \quad \text{Area} = \frac{1}{2} \times 4 \times 4 \times \sin 60^\circ$$

$$= 8 \times \frac{\sqrt{3}}{2}$$

$$= 4\sqrt{3} \text{ units}^2$$



$$(i) \quad \angle B = \angle C = 90^\circ \text{ angles in square}$$

$$BQ = CR \text{ given}$$

$$BP = QC \text{ subtraction of equal length from sides of square}$$

$$\therefore \underline{\triangle PBQ \cong \triangle QCR \text{ SAS}}$$

$$(ii) \quad \angle BQP = \angle QRC = x^\circ \text{ corresp. } \angle \text{s in congruent } \triangle \text{s,}$$

$$\angle CQR = 90^\circ - x^\circ \text{ } \angle \text{ in } \triangle QCR$$

$$\angle PQR = 180^\circ - (x^\circ + 90^\circ - x^\circ) \text{ st. } \angle \text{ BQC}$$

$$= 90^\circ$$

$$(c) \quad 2\sin^2 x = \sin x$$

$$2\sin^2 x - \sin x = 0$$

$$\sin x (2\sin x - 1) = 0$$

$$\sin x = 0 \quad \sin x = \frac{1}{2}$$

$$\underline{x = 0^\circ, 180^\circ, 360^\circ} \text{ or } \underline{x = 30^\circ, 150^\circ}$$

$$(d) \quad (i) \quad \alpha + \beta = -\frac{b}{a}$$

$$= -\frac{-8}{-1}$$

$$= -8$$

$$(ii) \quad \alpha\beta = \frac{c}{a}$$

$$= \frac{4}{-1}$$

$$= -4$$

$$(iii) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (-8)^2 - 2(-4)$$

$$= 72$$

$$(iv) \quad (\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$$

$$= 72 - 2(-4)$$

$$= 80$$

### Question 10

$$(a) \quad 4 + 23x^2 - 6x^4 = 0$$

$$(4 - x^2)(1 + 6x^2) = 0$$

$$4 - x^2 = 0 \quad 1 + 6x^2 = 0$$

$$x^2 = 4 \quad x^2 = -\frac{1}{6}$$

$$x = \pm 2 \quad \text{no solution}$$

$$(b) \quad f(x) = 2x - x^3$$

$$f(-x) = 2(-x) - (-x)^3$$

$$= -2x + x^3$$

$$= -(2x - x^3)$$

$$= -f(x)$$

$$f(-x) = -f(x) \therefore \underline{f(x) \text{ is odd}}$$

$$(c) \quad x^2 \equiv A(2-x)^2 + B(2-x) + C$$

$$= A(4 - 4x + x^2) + 2B - Bx + C$$

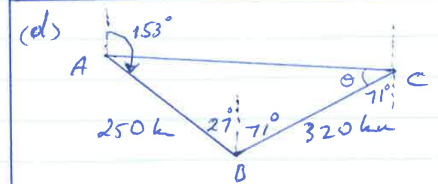
$$= Ax^2 + (-4A - B)x + (4A + 2B + C)$$

$$\underline{A=1} \quad -4A - B = 0 \quad 4A + 2B + C = 0$$

$$-4 - B = 0 \quad 4 - 8 + C = 0$$

$$\underline{B=-4} \quad \underline{C=4}$$

$$\therefore \underline{x^2 \equiv (2-x)^2 - 4(2-x) + 4}$$



$$(i) \quad AC^2 = 250^2 + 320^2 - 2 \times 250 \times 320 \cos 98^\circ$$

$$= 187167.6962$$

$$\underline{AC \doteq 433 \text{ km}}$$

$$(ii) \quad \frac{\sin \theta}{250} = \frac{\sin 98^\circ}{433}$$

$$\sin \theta = \frac{\sin 98^\circ}{433} \times 250$$

$$\sin \theta = 0.5717 \dots$$

$$\underline{\theta \doteq 35^\circ}$$

$$\therefore \text{bearing} = 190^\circ + 71^\circ + 35^\circ$$

$$= \underline{296^\circ}$$

$$(e) \quad \frac{d}{dx} \left( \frac{4x^2}{\sqrt{3x-2}} \right) = \frac{(3x-2)^{\frac{1}{2}} \cdot 8x - 4x^2 \cdot \frac{1}{2}(3x-2)^{-\frac{1}{2}}}{3x-2}$$

$$= \frac{(3x-2) \cdot 8x - 6x^2}{(3x-2)^{\frac{3}{2}}(3x-2)}$$

$$= \underline{\underline{\frac{18x^2 - 16x}{(3x-2)^{\frac{3}{2}}}}}$$